what is a qubit ?!

classical bit

unit of information
classical computer

bit 0

bit 1

classical bit

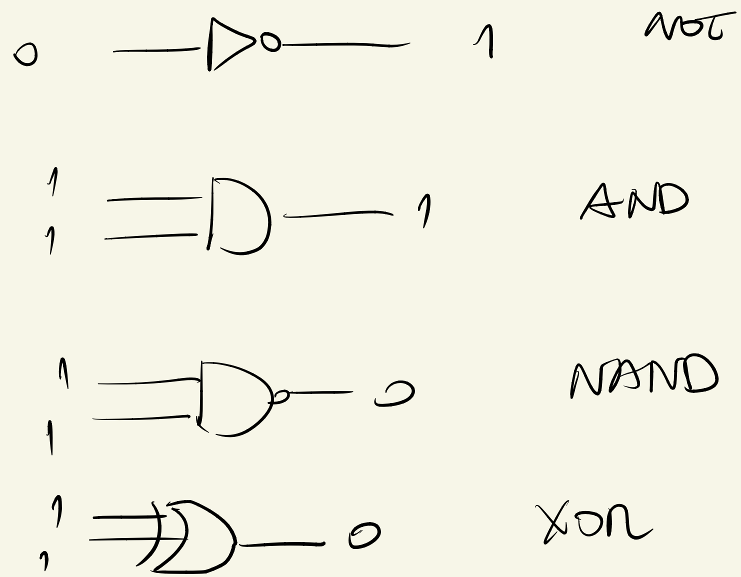
unit of information
classical computer

bit 0

bit 1

How do we manipulate bits?

logical gates



universal gate set for
classical computation
(only NANDS actually)

classical bit

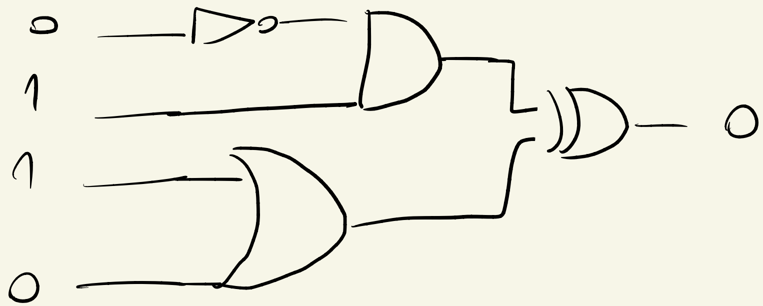
unit of information
classical computer

bit 0

bit 1

How do we manipulate bits ?

logical gates



Deterministic
computation

classical bit

unit of information
classical computer

bit 0

bit 1

(deterministic)

Quantum bit

unit of information
~~classical~~ computer
Quantum

qubit $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

qubit $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

but

classical bit

unit of information
classical computer

bit 0

bit 1

Quantum bit

unit of information
~~classical~~ computer
Quantum

qubit $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

qubit $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

but

qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$= \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

linear combination

The outcome of the quantum computation is

Probabilistic

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The outcome of the quantum computation is probabilistic

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Does it mean the qubit is a probabilistic bit?

The outcome of the quantum computation is probabilistic

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Does it mean the qubit is a probabilistic bit?

No, because $\alpha, \beta \in \mathbb{C}$

Amplitudes can interfere

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha, \beta \in \mathbb{C}$$

$\longrightarrow$ Probabity of obtaining 0,1

$$P_{|0\rangle} = |\alpha|^2$$

$$P_{|1\rangle} = |\beta|^2$$

$$P_{|0\rangle} + P_{|1\rangle} = 1$$

Always?

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ $\longrightarrow$ Probabity of obtaining 0,1

$\alpha, \beta \in \mathbb{C}$

$P_{|0\rangle} = |\alpha|^2$

$P_{|1\rangle} = |\beta|^2$

$P_{|0\rangle} + P_{|1\rangle} = 1$

Always ?

The qubit is a complex vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ that by the end of the computation returns a classical bit w/ probabilities $|\alpha|^2, |\beta|^2$ s.t

$$|\alpha|^2 + |\beta|^2 = 1$$

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \longrightarrow$ Probabity of obtaining 0,1

$\alpha, \beta \in \mathbb{C}$

$$P_{|0\rangle} = |\alpha|^2$$

$$P_{|1\rangle} = |\beta|^2$$

$$P_{|0\rangle} + P_{|1\rangle} = 1$$

Always?

How do we manipulate the qubit?

Quantum logic gates that...

Preserve the length of the vector $|\alpha|^2 + |\beta|^2 = 1$

Preserve the length of the vector $|\alpha|^2 + |\beta|^2 = 1$

unitary matrices act as our logic gates

$$U^{+} U = I$$

Preserve the length of the vector $|\alpha|^2 + |\beta|^2 = 1$

unitary matrices act as our logic gates

$$U^\dagger U = I \qquad U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad U^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

quantum gates $\longrightarrow$ quantum circuits

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \qquad |0\rangle \,—\, \boxed{U} \,—$$

$$|1\rangle \,—\, \boxed{U} \,—$$

But crucially quantum gates can receive superpositions as input

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad \rightsquigarrow \quad |\psi\rangle \text{ --- } \boxed{U} \text{ --- } |\psi'\rangle$$

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

U is a linear operator
(acts in each term of the superposition)

$$|\psi'\rangle = U|\psi\rangle =$$

$$= U\begin{pmatrix} \alpha \\ \beta \end{pmatrix} =$$

$$= U\begin{pmatrix} \alpha \\ 0 \end{pmatrix} + U\begin{pmatrix} 0 \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$$

Ex: Verify if the following matrices are valid quantum gates

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = A^\dagger \qquad AA^\dagger = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = B^\dagger \qquad BB^\dagger = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = C^\dagger \longrightarrow CC^\dagger = \underline{I}$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$-\boxed{X}- \qquad -\boxed{Y}- \qquad -\boxed{Z}-$$

Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$-\boxed{X}- \qquad\qquad -\boxed{Y}- \qquad\qquad -\boxed{Z}-$$

Pauli matrices

Ex: Verify what they do to our computational basis states $|0\rangle, |1\rangle$

$X|0\rangle = |1\rangle$
$X|1\rangle = |0\rangle$ $\rightsquigarrow$ Not gate

$y|0\rangle = i|1\rangle$
$\rightsquigarrow$ Complex NOT gate
$y|1\rangle = -i|0\rangle$

$z|0\rangle = |0\rangle$ $\rightsquigarrow$ phase shift $|1\rangle$
$z|1\rangle = -|1\rangle$

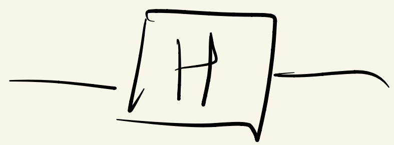
what about $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$

$$H = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} = H^{\dagger} \rightarrow HH^{\dagger} = \mathbb{1}$$

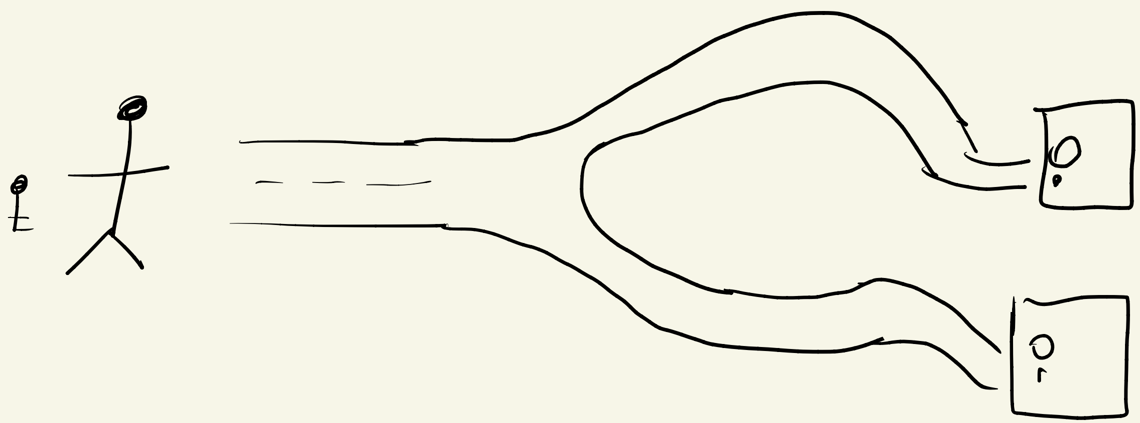
$$H|0\rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$H|1\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Hadamard gate
creates superposition



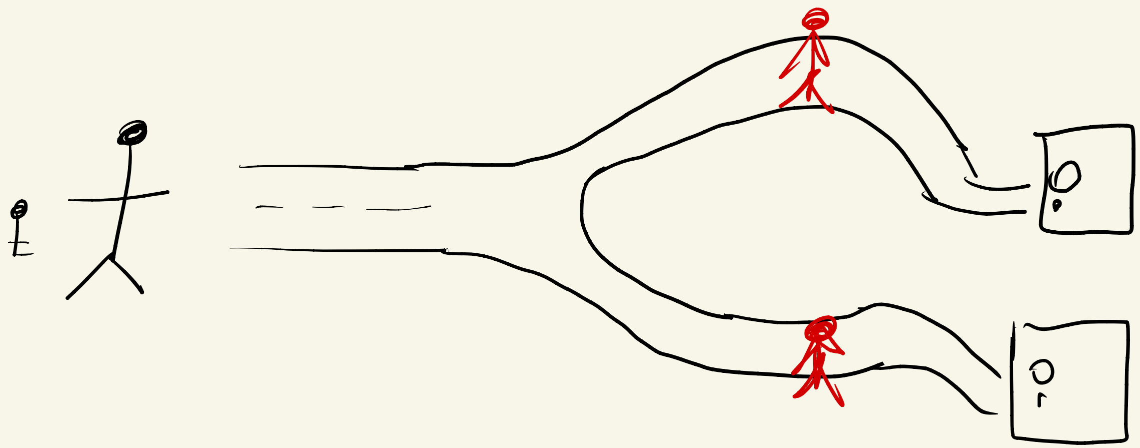
Imagine your favorite game character



Road 0, 1

classical bot would test
one road at a time

Imagine your favorite game character



Hadamards create superpositions so the quantum bot experiences both roads

Awesome! Is it enough?

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Quantum computation would return the correct road w/ probability

1/2 !

So, superposition alone is not enough !

Ex:

$$|0\rangle \;-\!\!-\boxed{H}-\!\!-\boxed{Z}-\!\!-\boxed{H}-\!\!-\quad ?$$

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$Z\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

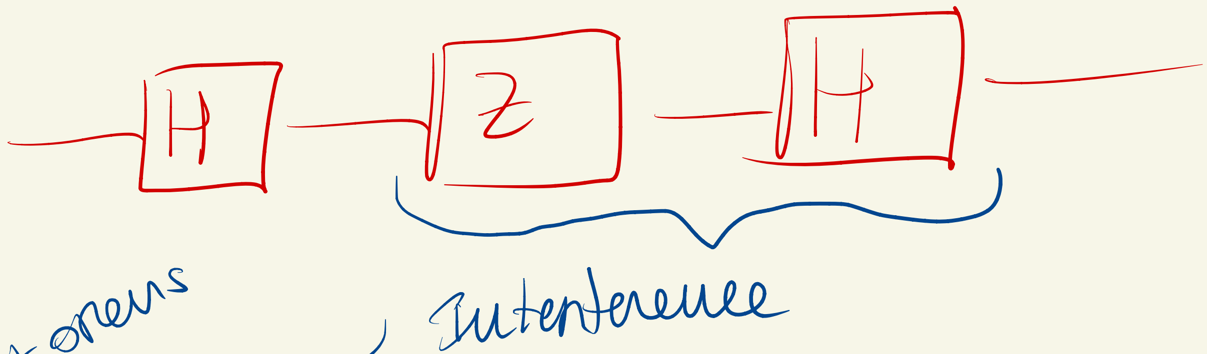
$$H\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right)$$

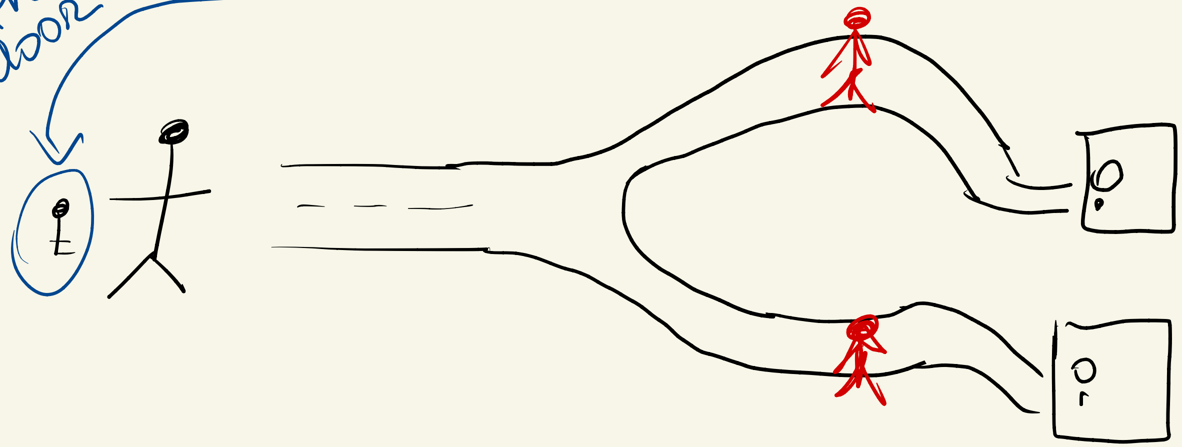
$$- \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right)$$

$$= |1\rangle$$

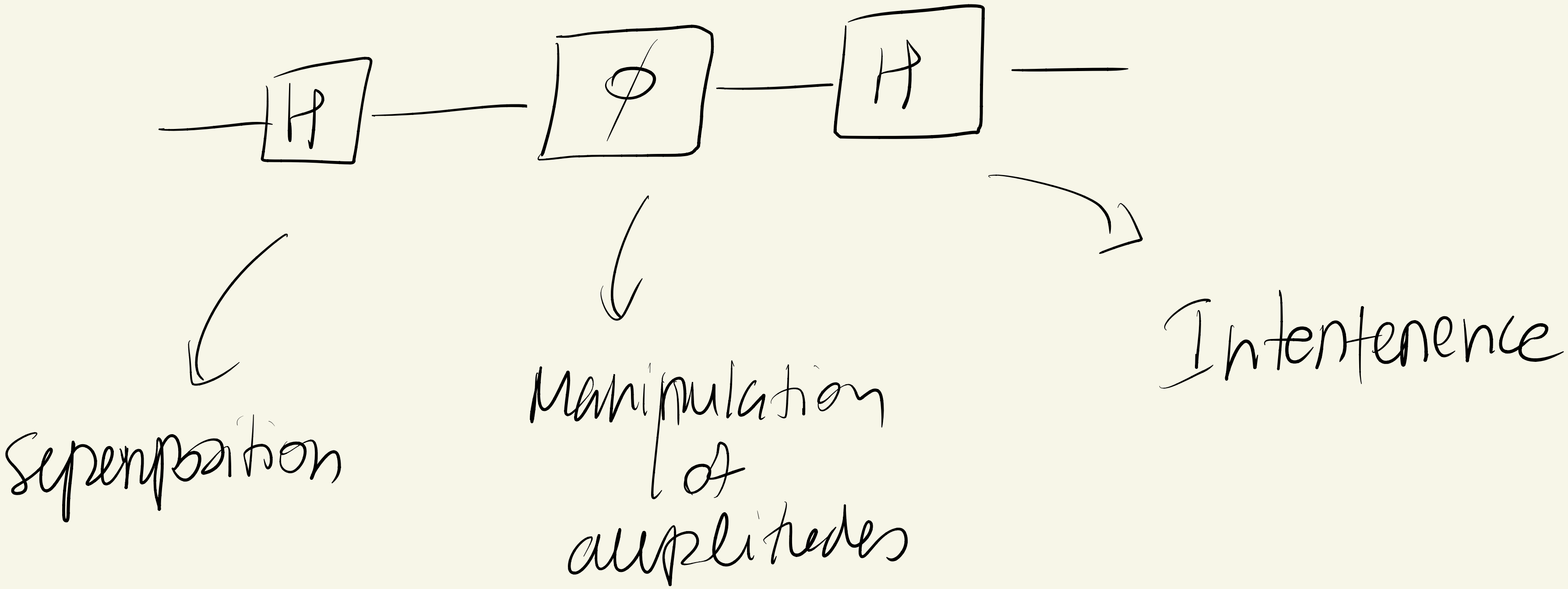
Amplitudes intentened s.t amplitude $|0\rangle$ is zero!



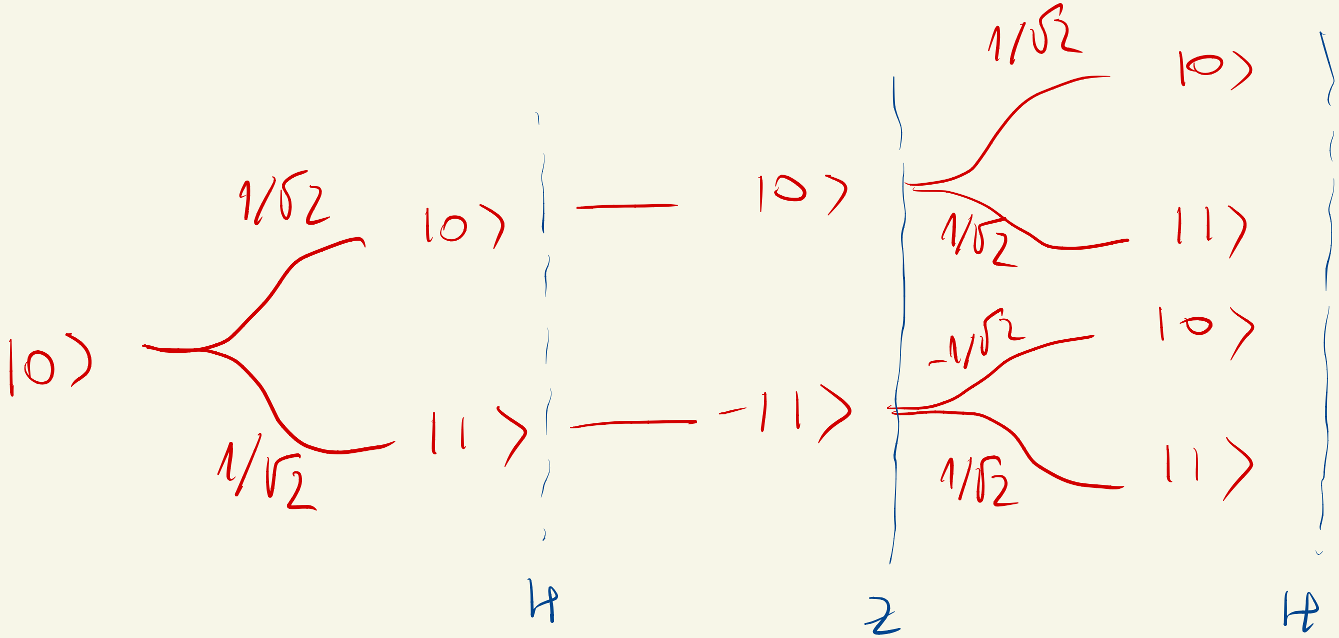
would be the key that opens the door.



A quantum algorithm will be in general a computation



$|0\rangle - \boxed{H} - \boxed{Z} - \boxed{H} -$



$$\alpha_0 = \frac{1}{2} - \frac{1}{2} = 0$$

$$\alpha_1 = \frac{1}{2} + \frac{1}{2} = 1$$

$|0\rangle - \boxed{H} - \boxed{P_\phi} - \boxed{H} -$

$$P_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \quad \rightsquigarrow \quad \phi = \pi \rightarrow e^{i\pi} = \cos(\pi) + i\,\mathrm{sen}(\pi)$$

$$= -1$$

$$\downarrow$$

$$- \boxed{P_\pi} - \; = \; - \boxed{Z} -$$

phase shift

$$|0\rangle - \boxed{H} - \boxed{P_\phi} - \boxed{H} -$$

$$P_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \rightsquigarrow \phi = \pi \rightarrow e^{i\pi} = \cos(\pi) + i\,\mathrm{sen}(\pi)$$

$$= -1$$

$$\downarrow$$

$$- \boxed{P_\pi} - \;=\; - \boxed{Z} -$$

Ex: what is the value of α_0, α_1 ?

and $P_0 = |\alpha_0|^2$

$P_1 = |\alpha_1|^2$

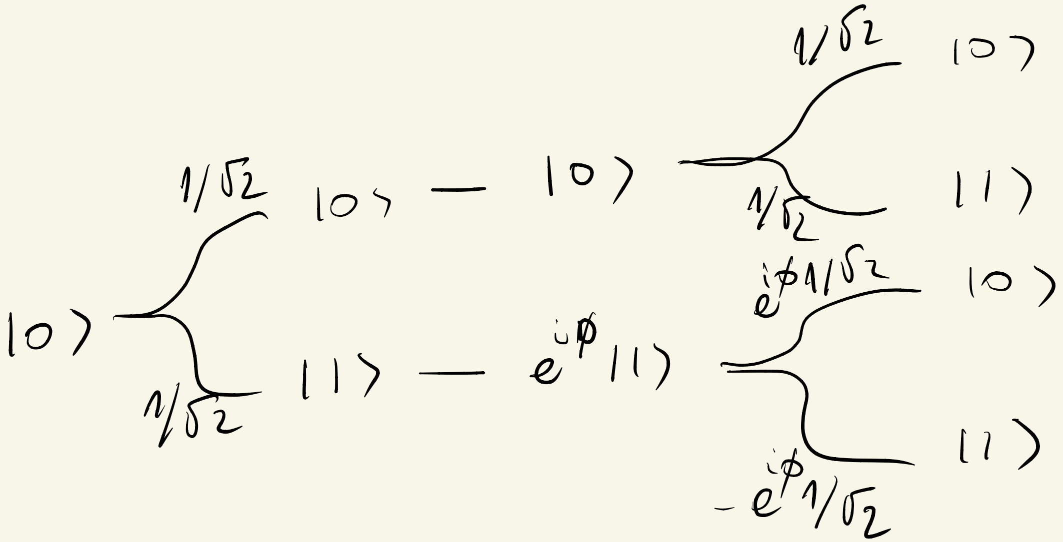
$$|0\rangle - \boxed{H} - \boxed{P_\phi} - \boxed{H} -$$

$$P_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \rightsquigarrow \phi = \pi \to e^{i\pi} = \cos(\pi) + i\,\mathrm{sen}(\pi)$$

$$= -1$$

$$\downarrow$$

$$-\boxed{P_\pi}- \; = \; -\boxed{Z}-$$



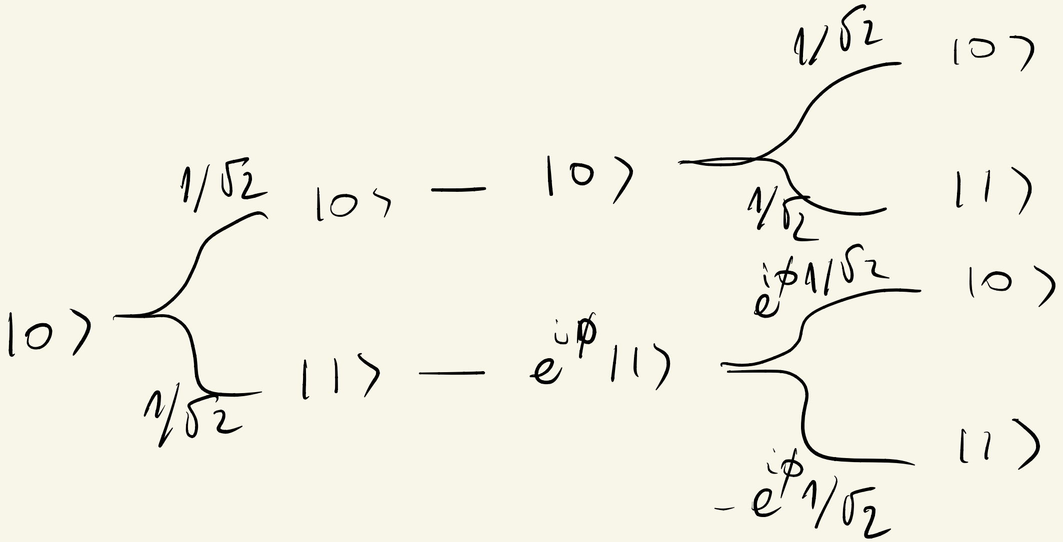
$|0\rangle - \boxed{H} - \boxed{P_\phi} - \boxed{H} -$

$$P_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \rightsquigarrow \quad \phi = \pi \rightarrow e^{i\pi} = \cos(\pi) + i\,\mathrm{sen}(\pi)$$

$$= -1$$

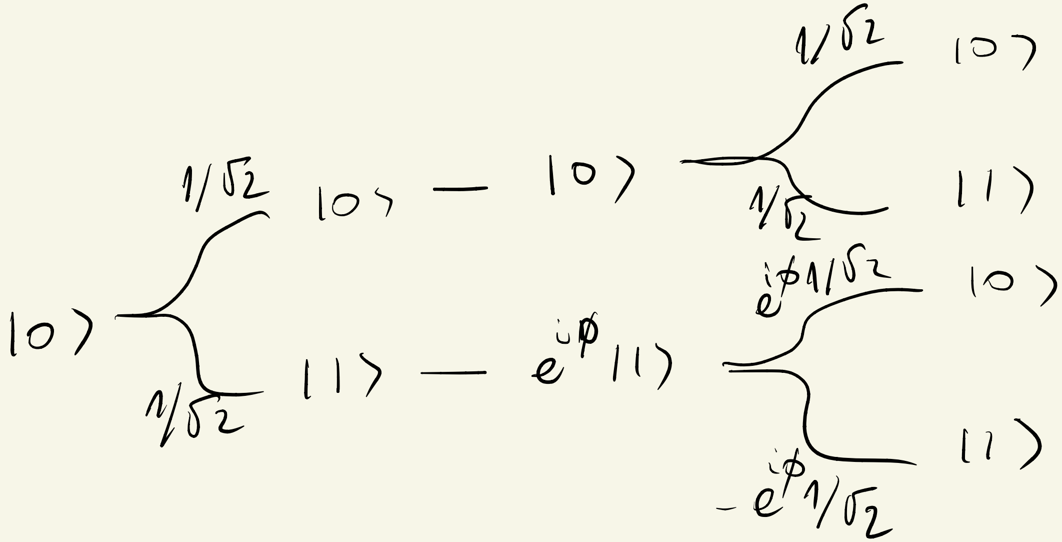
$$\downarrow$$

$$-\boxed{P_\pi}- \; = \; -\boxed{Z}-$$



$$\alpha_0 = \frac{1}{2} + \frac{1}{2} e^{i\phi}$$

$$\alpha_1 = \frac{1}{2} - \frac{1}{2} e^{i\phi}$$



$$\alpha_0 = \frac{1}{2} + \frac{1}{2} e^{i\phi}$$

$$\alpha_1 = \frac{1}{2} - \frac{1}{2} e^{i\phi}$$

$$P_0 = |\alpha_0|^2 = \alpha_0 \alpha_0^* = \left(\frac{1}{2} + \frac{1}{2} e^{i\phi}\right)\left(\frac{1}{2} + \frac{1}{2} e^{-i\phi}\right)$$

$$= \frac{1}{4} + \frac{1}{4} e^{-i\phi} + \frac{1}{4} e^{i\phi} + \frac{1}{4}$$

$$= \frac{1}{2} + \frac{1}{4}\left(e^{i\phi} + e^{-i\phi}\right)$$

$$= \frac{1}{2} + \frac{1}{2}\cos(\phi)$$

gate set so far

$\{ H, X, Y, Z, \phi \}$

Ex : Is the following matrix a valid quantum logic gate ?

$$U = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$UU^{\dagger} = I$$

$$\begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\operatorname{sen}\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & \operatorname{sen}\left(\frac{\theta}{2}\right) \\ -\operatorname{sen}\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2\left(\frac{\theta}{2}\right) + \operatorname{sen}^2\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right)\operatorname{sen}\left(\frac{\theta}{2}\right) - \operatorname{sen}\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) - \cos\left(\frac{\theta}{2}\right)\operatorname{sen}\left(\frac{\theta}{2}\right) & \operatorname{sen}^2\left(\frac{\theta}{2}\right) + \cos^2(\theta) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Ex : what does it do to $|0\rangle, |1\rangle$?

$$U = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\operatorname{sen}\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$R_y(\theta)\,|0\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\operatorname{sen}\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos(\theta/2) \\ \operatorname{sen}(\theta/2) \end{pmatrix}$$

$$= \cos\left(\frac{\theta}{2}\right)|0\rangle + \operatorname{sen}\left(\frac{\theta}{2}\right)|1\rangle$$

$$R_y(\theta)\,|1\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\operatorname{sen}\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\operatorname{sen}\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$= -\operatorname{sen}\left(\frac{\theta}{2}\right)|0\rangle + \cos\left(\frac{\theta}{2}\right)|1\rangle$$

Intermission

Ex : Is the following quantum state valid?

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5} e^{\pi/6 i} |1\rangle$$

$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6 i}|1\rangle$$

$$P_0 + P_1 = 1$$

$$P_0 = |\alpha_0|^2 = \frac{9}{25}$$

$$P_1 = |\alpha_1|^2 = \alpha_1 \alpha_1^*$$

$$= -\frac{4}{5}e^{\pi/6 i} \cdot -\frac{4}{5}e^{-\pi/6 i}$$

$$= \frac{16}{25}$$

$$P_0 + P_1 = 1 \quad \checkmark$$

$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6 i}|1\rangle$$

$$P_0 + P_1 = 1 \qquad P_0 = |\alpha_0|^2 = \frac{9}{25} \qquad P_1 = |\alpha_1|^2 = \alpha_1\alpha_1^*$$

$$= -\frac{4}{5}e^{\pi/6 i} \cdot -\frac{4}{5}e^{-\pi/6 i}$$

$$= \frac{16}{25}$$

$$P_0 + P_1 = 1$$

ou

$$\langle\psi|\psi\rangle = \left(\frac{3}{5}\langle 0| - \frac{4}{5}e^{-\pi/6 i}\langle 1|\right)\left(\frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6 i}|1\rangle\right)$$

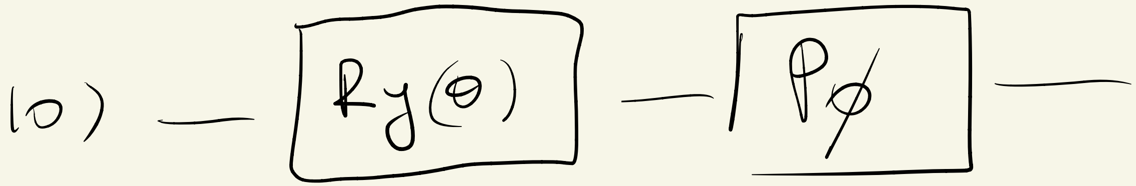
$$\langle 0|0\rangle = 1 \qquad = \frac{9}{25} + \frac{16}{25} = 1$$

$$\langle 0|1\rangle = 0$$

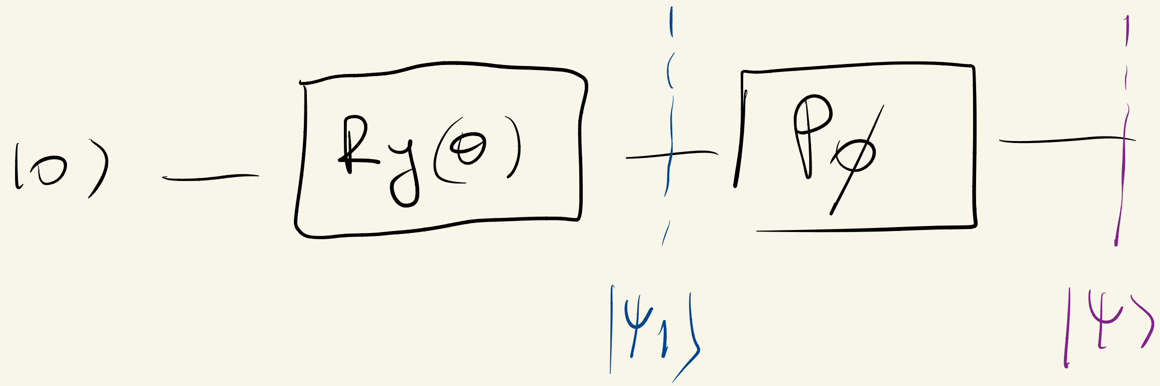
Ex: Implement the circuit to prepare $|\psi\rangle$

$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6 i}|1\rangle$$

$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6\, i}|1\rangle$$



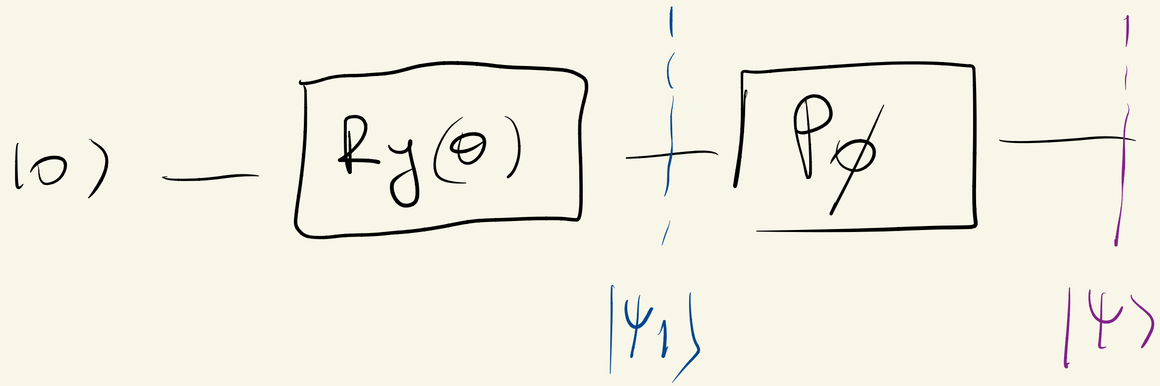
$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6\, i}|1\rangle$$



$$Ry(\theta)|0\rangle = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) \end{pmatrix} = \cos\left(\frac{\theta}{2}\right)|0\rangle + \sin\left(\frac{\theta}{2}\right)|1\rangle$$

$$= |\psi_1\rangle$$

$$|\psi\rangle = \frac{3}{5}|0\rangle - \frac{4}{5}e^{\pi/6\, i}|1\rangle$$



$$|\psi\rangle = P_\phi\,|\psi_1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \operatorname{sen}\left(\frac{\theta}{2}\right) \end{pmatrix} = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi}\operatorname{sen}\left(\frac{\theta}{2}\right) \end{pmatrix}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + \operatorname{sen}\left(\frac{\theta}{2}\right) e^{i\phi}|1\rangle$$

$$\cos^2\left(\frac{\theta}{2}\right) = \frac{9}{25} \longrightarrow \theta = 2\arccos\left(\sqrt{\frac{9}{25}}\right)$$

$$\operatorname{sen}^2\left(\frac{\theta}{2}\right) = 1 - \cos^2\left(\frac{\theta}{2}\right) = 1 - \frac{9}{25} = \frac{16}{25} \quad ↯$$

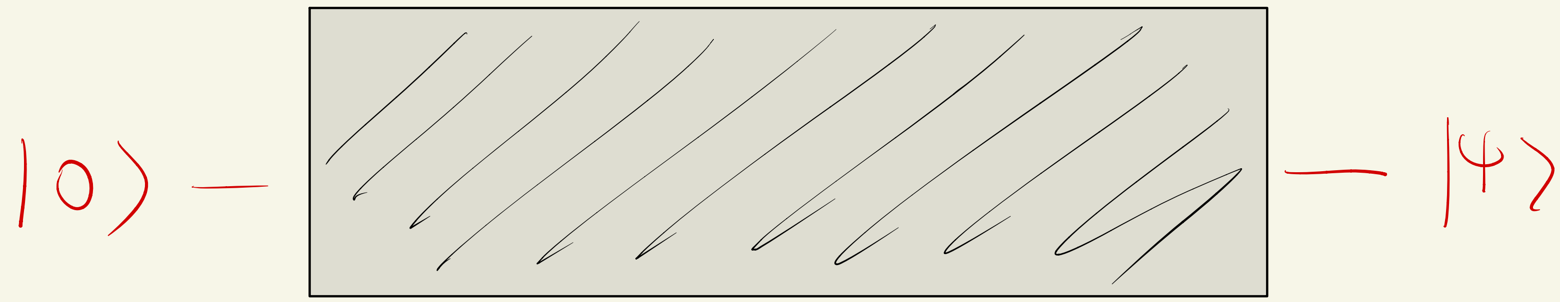
$$e^{i\phi} = -e^{i\pi/6} \rightsquigarrow \phi = \pi + \pi/6$$

$$e^{(\pi+\pi/6)i} = e^{\pi i} e^{\pi/6\, i} = -e^{\pi/6\, i}$$

$$|0\rangle \;—\; \boxed{R_y\left(2\arccos\sqrt{\tfrac{9}{25}}\right)} \;—\; \boxed{P_{\pi+\pi/6}} \;—\; |\psi\rangle$$

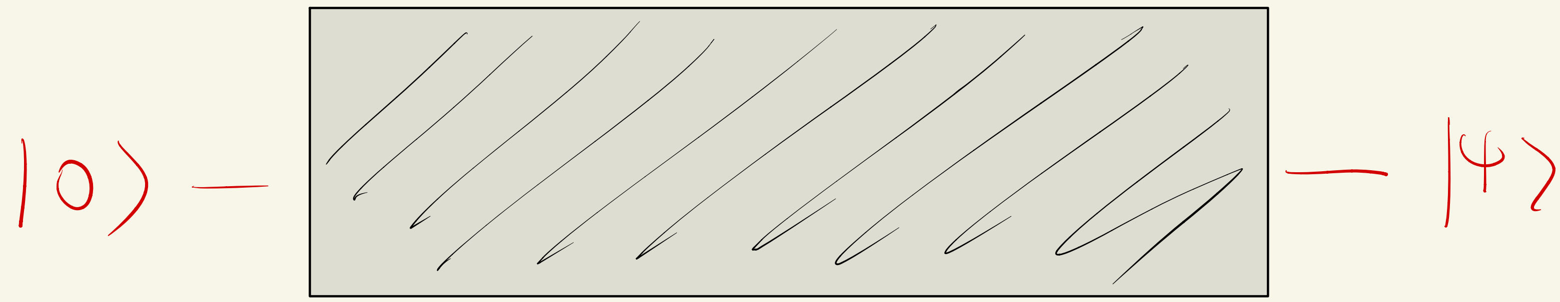
Quantum computation terminates

in state $\begin{cases} |0\rangle & \text{w/ Prob } 9/25 \\ |1\rangle & \text{w/ Prob } 16/25 \end{cases}$



Imagine we're given $|\psi\rangle$ as a blackbox ?

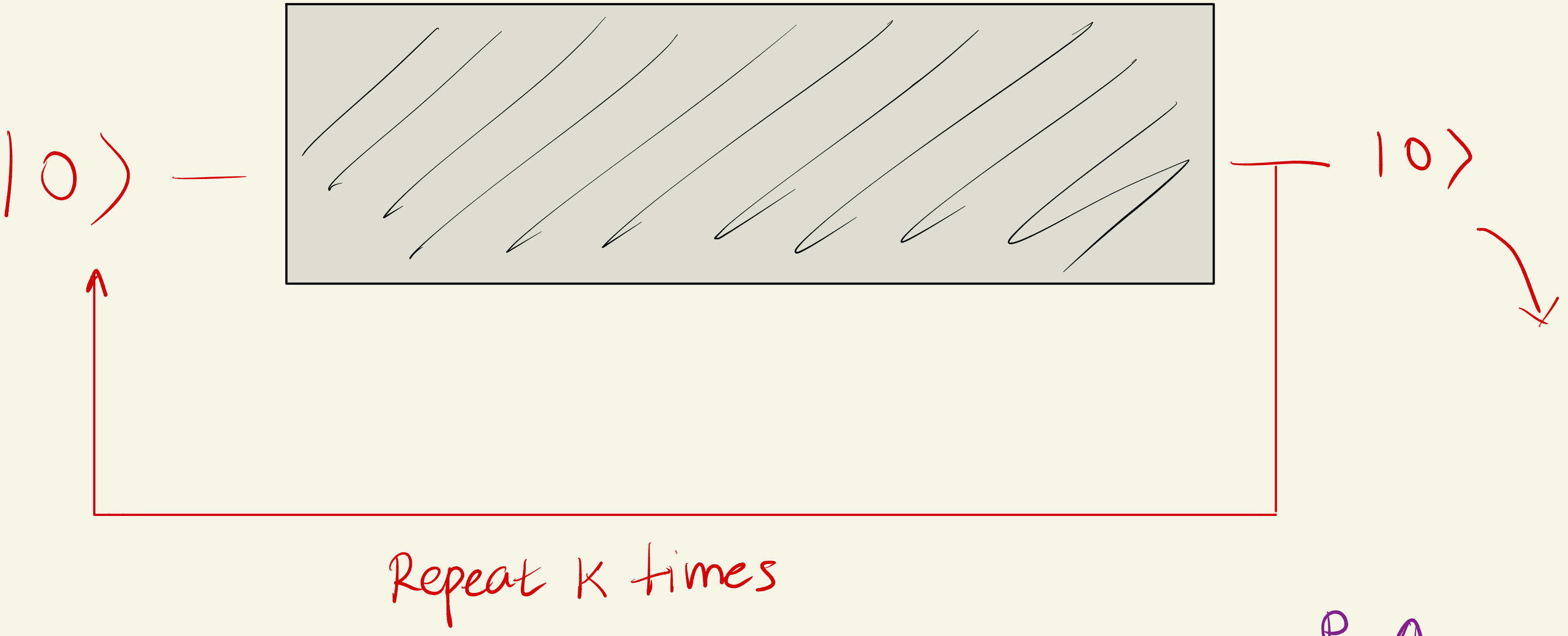
How can we know the value of P_0, P_1 ?

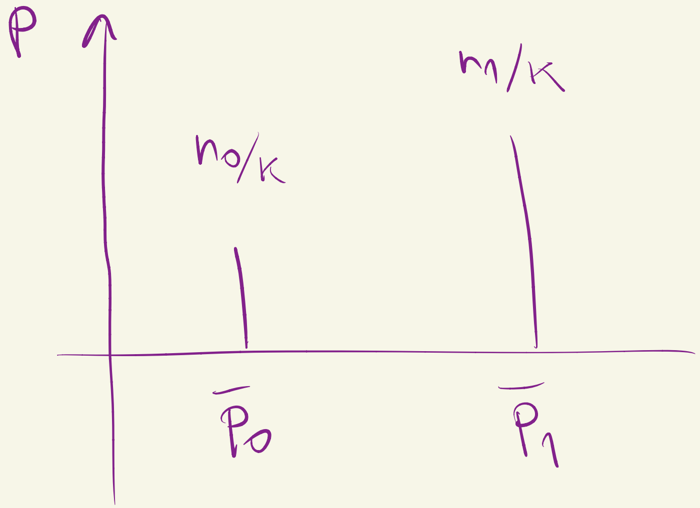


Imagine we're given $|\psi\rangle$ as a blackbox?

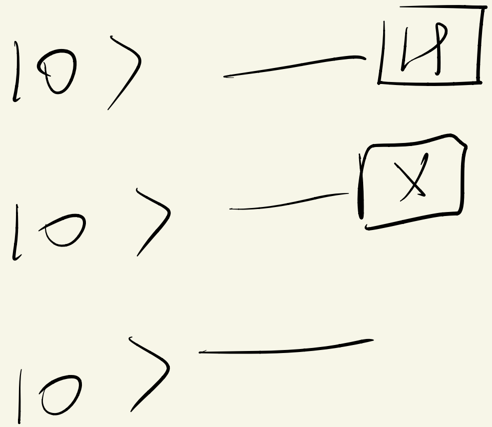
How can we know the value of p_0, p_1?

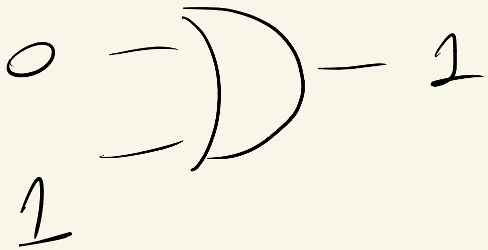
MAKE QUERIES TO THE BLACKBOX!





Law of large numbers $K \to \infty$ $\bar{P}_0 \approx P_0$





There is certainly many more single qubit gates but multiple qubit computation require multiqubit gates